Mathematical Typesetting in LaTeX

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1 Inline and Display Mathematics

Inline mathematics appears within a sentence, for example $x^2 + y^2 = z^2$.

Display mathematics appears on a separate line:

$$x^2 + y^2 = z^2$$

2 Superscripts and Subscripts

$$a_i^2 + b_j^3$$

$$x_{ij}, \quad x^{n+1}, \quad a_{i,j}^2$$

3 Greek Letters and Mathematical Symbols

Greek letters:

$$\alpha,\ \beta,\ \gamma,\ \delta,\ \lambda,\ \pi,\ \sigma$$

Relations and set symbols:

$$a \leq b, \quad x \neq y, \quad A \subseteq B, \quad A \cup B, \quad A \cap B$$

Binary operators:

$$a \times b, \quad A \otimes B, \quad A \oplus B$$

4 Fractions and Roots

$$\frac{a+b}{c}, \qquad \sqrt{x}, \qquad \sqrt[3]{y}$$

5 Mathematical Fonts

Different mathematical fonts can be used to represent expressions.

$$\mathcal{ABC} \qquad \mathfrak{ABC} \qquad \mathrm{ABC}$$

$$ABC \qquad \mathbf{ABC} \qquad \mathsf{ABC}$$

6 Mathematical Display Styles

The same expression can appear in different mathematical styles.

$$\sum_{i=1}^{n} i$$

$$\sum_{i=1}^{n} i$$

$$\textstyle\sum_{i=1}^{n} i$$

$$\scriptstyle\sum_{i=1}^{n} i$$

$$\scriptscriptstyle\sum_{i=1}^{n} i$$

7 Summation, Product and Integration

Summation:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Product:

$$\prod_{k=1}^{m} k$$

Definite integral:

$$\int_0^1 x^2\,dx = \frac{1}{3}$$

Other integral symbols:

$$\oint_C f(z)\,dz$$

8 Numbered Equations

The equation environment automatically numbers equations.

$$E = mc^2 \tag{1}$$

$$\int_0^1 x^2 \, dx = \frac{1}{3} \tag{2}$$

9 Aligned Equations

The align environment aligns multiple equations at a common point.

$$a + b = c \tag{3}$$
$$x + y = 10 \tag{4}$$
$$2x + 3y = 15 \tag{5}$$

10 Arrays

The array environment creates columns with different alignment options.

Left	Center	Right
a	b	c
1	2	3

11 Matrices

11.1 Basic Matrix

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix}$$

11.2 Matrix with Parentheses

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

11.3 Matrix with Square Brackets

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

11.4 Matrix with Curly Braces

$$\begin{Bmatrix} x & y \\ z & w \end{Bmatrix}$$

11.5 Determinant

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

11.6 Double Vertical Bars

$$\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$$

12 Description Lists

The description environment creates lists containing terms and their explanations.

Matrix A rectangular arrangement of numbers in rows and columns.

Determinant A scalar value associated with a square matrix.

Integral A mathematical operation used to calculate areas and accumulated quantities.